

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Wed Jul 22 01:48:40 2026
| Host : AK_LAPTOP running 64-bit major release (build
9200)
| Command : report_utilization -file
uart_top_utilization_synth.rpt -pb uart_top_utilization_synth.pb
| Design : uart_top
| Device : xc7a15tcpvg236-1
| Speed File : -1
Design State : Synthesized

Utilization Design Information

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1. Slice Logic
-

Site Type	Used	Fixed	Prohibited	Available
Util%				
Slice LUTs*	50	0	0	10400
0.48				
LUT as Logic	50	0	0	10400
0.48				
LUT as Memory	0	0	0	9600
0.00				
Slice Registers	60	0	0	20800
0.29				
Register as Flip Flop	60	0	0	20800
0.29				
Register as Latch	0	0	0	20800
0.00				
F7 Muxes	1	0	0	16300
<0.01				
F8 Muxes	0	0	0	8150
0.00				
Unique Control Sets	9		0	8150
0.11				

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	—	—
0	—	—	Set
0	—	—	Reset
0	—	Set	—
0	—	Reset	—
0	Yes	—	—
0	Yes	—	Set
0	Yes	—	Reset
1	Yes	Set	—
59	Yes	Reset	—

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	25	0.00
RAMB36/FIFO*	0	0	0	25	0.00
RAMB18	0	0	0	50	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	45	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	25	0	0	106	23.58
Bonded IPADs	0	0	0	10	0.00
Bonded OPADs	0	0	0	4	0.00
PHY_CONTROL	0	0	0	5	0.00
PHASER_REF	0	0	0	5	0.00
OUT_FIFO	0	0	0	20	0.00
IN_FIFO	0	0	0	20	0.00
IDELAYCTRL	0	0	0	5	0.00
IBUFDS	0	0	0	104	0.00
GTPE2_CHANNEL	0	0	0	2	0.00

PHASER_OUT/PHASER_OUT_PHY		0		0		0	
20		0.00					
PHASER_IN/PHASER_IN_PHY		0		0		0	
20		0.00					
IDELAYE2/IDELAYE2_FINEDELAY		0		0		0	
250		0.00					
IBUFDS_GTE2		0		0		0	
2		0.00					
ILOGIC		0		0		0	
106		0.00					
OLOGIC		0		0		0	
106		0.00					
+-----+-----+-----+-----+-----+							
---+-----+							

5. Clocking

+-----+-----+-----+-----+-----+-----+							
Site Type		Used		Fixed		Prohibited	
Available		Util%					
+-----+-----+-----+-----+-----+-----+							
BUFGCTRL		1		0		0	
BUFGCTRL		1		0		32	
BUFGCTRL		1		0		3.13	
BUFIO		0		0		0	
BUFIO		0		0		20	
BUFIO		0		0		0.00	
MMCME2_ADV		0		0		5	
MMCME2_ADV		0		0		0.00	
PLLE2_ADV		0		0		5	
PLLE2_ADV		0		0		0.00	
BUFMRCE		0		0		0	
BUFMRCE		0		0		10	
BUFMRCE		0		0		0.00	
BUFHCE		0		0		0	
BUFHCE		0		0		72	
BUFHCE		0		0		0.00	
BUFR		0		0		0	
BUFR		0		0		20	
BUFR		0		0		0.00	
+-----+-----+-----+-----+-----+-----+							

6. Specific Feature

+-----+-----+-----+-----+-----+-----+							
Site Type		Used		Fixed		Prohibited	
Available		Util%					

BSCANE2	0	0	0	4	0.00		
CAPTUREE2	0	0	0	1	0.00		
DNA_PORT	0	0	0	1	0.00		
EFUSE_USR	0	0	0	1	0.00		
FRAME_ECCE2	0	0	0	1	0.00		
ICAPE2	0	0	0	2	0.00		
PCIE_2_1	0	0	0	1	0.00		
STARTUPE2	0	0	0	1	0.00		
XADC	0	0	0	1	0.00		

7. Primitives

Ref Name	Used	Functional Category
FDRE	59	Flop & Latch
LUT6	19	LUT
LUT4	19	LUT
OBUF	14	IO
IBUF	11	IO
LUT5	9	LUT
LUT2	7	LUT
LUT3	5	LUT
CARRY4	4	CarryLogic
MUXF7	1	MuxFx
LUT1	1	LUT
FDSE	1	Flop & Latch
BUFG	1	Clock

8. Black Boxes

+	-----	+
	Ref Name	Used
+	-----	+

9. Instantiated Netlists

+	-----	+
	Ref Name	Used
+	-----	+